Report

Optimizing the U.S. Energy Mix to Reduce Greenhouse Gas Emissions by 2030

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Introduction

Topic

For this project, I'm investigating how energy consumption affects greenhouse gas (GHG) emissions. As stated by the United Nations (2025), a significant cause of climate change is the global emissions that come from generating electricity and heat through the combustion of fossil fuels. Furthermore, fossil fuels (coal, oil, and gas) account for more than 75% of global greenhouse gas emissions, making them the primary driver of climate change. Climate change encompasses various effects beyond global warming, including intensified storms, increased health risks, and rising oceans. Given these severe impacts of climate change, international initiatives such as the Paris Agreement have set ambitious targets, including achieving net-zero emissions by 2050 (IEA, 2021). According to Barboza (2021), U.S. commitment to addressing climate change has fluctuated with changes in the governing political party. Upon rejoining the Paris Agreement in 2021, the United States committed to a 50-52% emissions reduction by 2030, relative to 2005 levels. As international efforts to fight climate change increase, it is vital to analyze the emissions impact of different energy sources and assess whether the U.S. can meet its promised emissions cuts by 2030.

Research Question

The research question for this project is: “What is the statistical relationship between energy source composition and greenhouse gas emission intensity in U.S., and what is the optimal energy mix that minimizes emissions by 2030 under projected demand trends?”

For the first part, I will determine the relationship between greenhouse gas emissions from energy and the distribution of energy sources, including coal, gas, oil, wind, solar, hydropower and nuclear power. For the second part, given these relationships and the emission

factors, I will find an energy mix that minimizes emissions while satisfying the forecasted energy demand for 2030. Although emissions forecasting and energy systems are well-researched, we still need to identify the optimal energy mix for minimizing emissions within realistic demand constraints. This project aims to bridge this gap by using data analysis and optimization to suggest a possible path toward energy restructuring.

Information

For this project, I'm combining two datasets: "Global Historical Emissions from 1850 to 2023" (Climate Watch, 2020) and "Global primary energy consumption by source from 1800 to 2023" (Ritchie & Rosado, 2020). The "Global Historical Emissions" table contains GHG emissions (in tCO_2e) for different countries and from different sectors, including energy, agriculture, waste, industrial processes and others, so I will extract the emissions only from the energy sector for my final dataset. The "Global primary energy consumption" table has all the data that I need, specifically the amount of energy (in TWh) from nine sources, including solar, wind, nuclear, coal, natural gas and others. I will have to extract the information related to United States, merge and clean these datasets to create a unified dataset for my analysis. Though these tables are very valuable, they do not contain all the necessary data to answer my whole question, only the first part.

To avoid extrapolating beyond the available data, for my second part of the question, I will be using a predicted energy demand for 2030 from the *Annual Energy Outlook 2025* report by the U.S. Energy Information Administration (EIA, 2025). The U.S. Energy Information Administration ran 11 cases to model a range of assumptions for the projected energy data. I will rely on the *Reference case*, which evaluates the U.S. energy markets' potential performance based on laws and regulations in effect as of December 2024, along with historical technological

growth trends, making its data more reliable than my own projections. According to this report, in the *Reference case* scenario, the projected total energy consumption will be 91.88 quads (quadrillion BTU), which is equivalent to 26,921 TWh.

I am also considering aiming to reduce the GHG emissions by 50% from 2005, as the U.S. pledged in 2021. The 2005 energy data in the dataset shows 6.32 $GtCO_2e$ (gigatons of carbon dioxide equivalent) of GHG emissions, which are direct emissions mainly resulting from combustion at power plants and end-use. It does not include other stages like extraction, storage, or transportation. To approximate full lifecycle emissions for the 2005 baseline, I applied a 35% uplift to the reported direct energy-sector emissions. The uplift is a simplifying modeling assumption intended to account for upstream and downstream stages such as extraction, construction, transportation, and disposal, which are included in lifecycle assessments (Sandoval, 2023). Reducing the baseline by 50% gives us a 4.265 $GtCO_2e$ cap for 2030, which will be the emissions cap for my optimization part.

Additionally, I found potentially critical information for the optimization model, namely, the life-cycle assessment (COWI, 2024) of each energy source, measured in gCO_2e/kWh (grams of carbon dioxide equivalent per kilowatt-hour) from extraction to disposal. For example, wind power emits approximately 11 gCO_2e/kWh, which accounts for emissions from raw material extraction, manufacturing, installation, maintenance, and end-of-life recycling. This is how much each type of power emits during its life cycle, according to this source: Hydropower – approximately 4 gCO_2e/kWh; Wind power – approximately 11 gCO_2e/kWh; Nuclear power – approximately 12 gCO_2e/kWh; Solar power – around 41 gCO_2e/kWh; Natural gas – 290-930 gCO_2e/kWh; Oil – 510-1170 gCO_2e/kWh; Coal – 740-1689 gCO_2e/kWh.

Assumptions and limitations

The model will estimate the 2030 energy demand using the EIA's *Reference case scenario* (26,921 TWh) and assume constant emissions intensity for each energy source. The model will also focus solely on gross emissions from energy consumption, excluding offsets, carbon sinks, and potential technological or policy changes that may alter emissions. The analysis will likely be restricted to the period 1965-2023 due to inconsistencies in earlier data.

Model

Selection

For the project, I will be using two models: Ridge regression (also called L2 Regularization) and Linear Programming (LP). The Ridge regression model is deterministic, relying on fixed relationships within historical data; similarly, linear programming (LP) is a deterministic optimization model, solving for the minimum with predefined parameters, excluding stochastic elements. The Ridge regression model will be used to determine the statistical relationship between greenhouse gas (GHG) emissions and the composition of U.S. energy sources. The Linear Programming model will be used to determine the optimal energy mix to minimize emissions by 2030 under projected energy demand.

I decided to use a regression model, since according to several sources, these types of models are used to determine the relationship between a dependent variable and one or more independent variables (Meerschaert, 2013; Sharma, 2022; Wohlwend, 2023). There are different types of regression models, including linear, logistic, polynomial, Ridge, Lasso, etc., (Sharma, 2022). Initially, I was planning to use a multiple linear regression model (MLR). My first attempts with this model, using all predictors, yielded a high R-squared (0.989) but unrealistic coefficients, such as a large positive beta for biofuels. Even after removing two predictors, the Python OLS output still flagged potential multicollinearity with the note: "The condition number

is large, 1.35e+04. This might indicate strong multicollinearity or other numerical problems." These results raised doubts about my model choice, prompting me to perform diagnostic tests, specifically checking for multicollinearity and validating the linear model's suitability.

The diagnostic tests supported the linearity assumption but confirmed strong multicollinearity. The Ramsey RESET test, which evaluates whether a linear model fails to capture nonlinear relationships (Ramchand, 2025), returned a p-value of 0.0689, above the 0.05 threshold, indicating no evidence of model misspecification. Additionally, the residuals-versus-predicted values scatterplot (Figure A1, Appendix A) showed a random distribution, further supporting the model's linearity. To check for multicollinearity, I used Variance Inflation Factor (VIF) analysis (Table A1, Appendix A), a statistical method that measures how much a predictor's variance is inflated due to correlations with other predictors (Potters, 2024). The VIF values were extremely high for most of the predictors, and a correlation matrix (Figure A2, Appendix A) confirmed these findings.

Given these results, I considered both Ridge and Lasso regression as an alternative model. I selected Ridge because, unlike Lasso—which can shrink some coefficients to zero and exclude predictors entirely (Wohlwend, 2023)—Ridge retains all included variables. Compared to a multiple linear regression model, Ridge regression modifies the objective function by minimizing the Residual Sum of Squares (RSS) (the total squared differences between observed and predicted emissions intensities) plus an L^2 penalty term. This penalty term reduces the effect of multicollinearity by preventing coefficients from becoming excessively large, trading a small increase in RSS for improved model stability and predictive performance. Considering the results of all the tests that were conducted, I concluded that using a Ridge regression model, with two predictors removed, would be the most suitable approach.

For the optimization part, I decided on a linear programming (LP) model to minimize emissions subject to constraints such as projected energy demand, emission reduction targets, and resource-specific limits. There are several optimization models, including unconstrained optimization, linear, nonlinear, convex, and multivariable (Meerschaert, 2013; Ye, n.d.). According to Basumallick (2022), LP is one of the most straightforward and computationally efficient optimization techniques, making it well suited for resource allocation problems with clearly defined linear objectives and constraints. In this context, the objective function is a weighted sum of life-cycle emission factors for each energy source, multiplied by their share in the mix, with constraints representing demand, minimum/maximum bounds for each source, and stability requirements. This formulation makes LP particularly effective for identifying the least-emissions configuration that still meets system needs. However, the model is limited by assumptions such as constant emission factors over time, perfect divisibility of energy resources, and the exclusion of nonlinear dynamics like storage losses or intermittency effects.

Creation

For the relationship analysis (the first part of the question), I created a Ridge regression model using historical U.S. data (1965-2023) from ourworldindata.org and climatewatchdata.org, with energy source percentages (Solar, Wind, Coal, etc.) as predictors and emission intensity (emissions from energy/total energy consumptions (tCO_2e/TWh)) as the response variable. The general form of a Ridge regression objective function is (IBM, 2023; Wohlwend, 2023):

$$\textbf{minimize}\,\boldsymbol{\beta}\left\{\underbrace{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}_{RSS}+\underbrace{\lambda\sum_{j=1}^{p}\beta_j^2}_{L2\ Penalty\ term}\right\}$$

where:

$$\hat{y}_i = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij};$$

- n is the number of observations (years in my dataset, 1965–2023, n = 59);
- p is the number of predictors (energy sources, p = 7);
- y_i is the emissions intensity (tCO_2e/Twh);
- $\hat{y}_i$ is the predicted emissions intensity for the observation i (emissions intensity in tCO_2e/Twh for year i);
- β_0 is the intercept term,
- x_{ij} represents the value of the jth predictor variable (energy source) for the ith observation (year).
- β_j are the regression coefficients corresponding to each energy source predictor x_{ij};
- λ is the regularization parameter controlling the penalty strength for large coefficients;

If I take into consideration my predictors: solar, wind, hydropower, nuclear, gas, oil, and coal, this becomes:

$$\textbf{minimize}\,\beta \left\{ \sum_{i=1}^{59} [y_i - (\beta_0 + \beta_1 \cdot solar\%_i + \beta_2 \cdot wind\%_i + \beta_3 \cdot hydropower\%_i + \beta_4 \cdot nuclear\%_i + \beta_5 \cdot gas\%_i + \beta_6 \cdot oil\%_i + \beta_7 \cdot coal\%_i)]^2 + \lambda(\beta_1{}^2 + \beta_2{}^2 + \beta_3{}^2 + \beta_4{}^2 + \beta_5{}^2 + \beta_6{}^2 + \beta_7{}^2) \right\}$$

To address the optimization aspect of my question, I developed a Linear Programming (LP) model designed to determine the optimal percentage allocation of each energy source, thereby minimizing greenhouse gas emissions by the year 2030, while simultaneously taking into account the projected total energy demand for that year. The general form of the LP model equation is (Basumallick,2022; Meerschaert, 2013):

$$\textbf{minimize}\, Z = E_{2030} \times \sum_{i=1}^{m} E_i x_i$$

where:

- Z are the total greenhouse gas emissions (tCO_2e);
- E_{2030} is the projected energy demand in TWh for 2030 (E_{2030}= 26921 TWh);
- m is the number of energy sources considered (m = 8);
- E_i are the Life Cycle Assessment (LCA) emission factors (tCO_2e/TWh) for source i;
- x_i are the percentages of each energy source.

The LP model with my data is:

$$\textbf{minimize } Z = 26921 \times (E_1x_1 + E_2x_2 + E_3x_3 + E_4x_4 + E_5x_5 + E_6x_6 + E_7x_7 + E_8x_8)$$

where:

- $Z \leq 4265000000$ tCO_2e,
- x_1 = solar energy,
- x_2 = wind energy,
- x_3 = hydropower energy,
- x_4 = nuclear energy,
- x_5 = natural gas energy,
- x_6 = oil energy,
- x_7 = coal energy,
- x_8 = other renewables energy,

- E_1 = LCA for solar energy (tCO_2e/TWh),
- E_2 = LCA for wind energy,
- E_3 = LCA for hydropower energy,
- E_4 = LCA for nuclear energy,
- E_5 = LCA for natural gas energy,
- E_6 = LCA for oil energy,
- E_7 = LCA for coal energy,
- E_8 = LCA for other renewables energy

The model is subject to these constraints:

- upper and lower bounds for each energy source, derived from historical Compound Annual Growth Rate (CAGR) and the Ridge regression results (Table D3, Appendix D);
- all the energy percentages summate to 100:

$$\sum x_i = 1;$$

- minimum base load, for stability and ensuring minimum amount of electricity (NPUC, 2023; United States Environmental Protection Agency, 2022):

$$x_2 + x_3 + x_4 + x_5 + x_7 \geq 50\%;$$

- maximum intermittency, due to limitations for solar and wind energy (NPUC, 2023):

$$x_1 + x_2 \leq 25\%$$

- fossil reduction, based on the Net-Zero Emissions by 2050 Paris Agreement (IEA, 2021):

$$x_5 + x_6 + x_7 \leq 55\%$$

If I run the LP model with these constraints, but without bounds, the model prioritizes the energy source that emits the least emissions, which, in our case is nuclear. Realistically, we can't go from 8% nuclear energy to 100% in eight years, so this is why these constraints are needed.

The information and data I have collected—such as historical GHG emissions data, energy source consumption percentages, projected energy demand for 2030, and life-cycle greenhouse gas emissions values are all incorporated into the model and used to address my research question. The dataset that contains historical data of the GHG emissions intensity and

percentage of energy from each source from 1965 to 2023 will manifest as input for the Ridge regression model (predictors/response) and inform constraints for the Linear Programming model (bounds from trends). For the Ridge regression model the data is sufficient to estimate relationships. For the Linear Programming model, I will use the projected energy demand for 2030 and the life-cycle greenhouse gas emissions values as coefficients. This information, along with the previously mentioned constraints and the bounds suggested by the Ridge regression results will be sufficient to find the optimal energy mix.

Process

To begin answering my question, I first cleaned and prepared the historical dataset. To utilize the data within my optimization model, a model that specifically requires data to be represented as percentages, I changed the data to percentages. To remove the impact on greenhouse gases caused by changing consumption patterns, I normalized emissions by total energy consumption for each year. Therefore, I included an emissions intensity column (tCO_2e/TWh), calculated by dividing the energy sector's annual GHG emissions (tCO_2e) by the total energy consumption (TWh) for each year. Additionally, I replaced all zero values with a small non-zero value, 0.0000001, as it improves the performance of certain calculations and models. I selected emissions intensity as the dependent variable, and the percentage shares of energy sources (coal, oil, gas, nuclear, wind etc.) as independent variables. I also experimented with various predictor combinations and ultimately removed “Other Renewables” and “Biofuels.” Using wind, hydropower, solar, nuclear, coal, natural gas, and oil as predictors yielded more realistic coefficients.

As I mentioned in the Selection section of this report, I first opted for a multiple linear regression model. After testing for linearity and multicollinearity, the results confirmed the

presence of the latter, prompting me to look for ways to address it. Several sources (Frost, 2017; *Regression with Variables Summing to 100% – Multicollinearity Issue? - Statalist*, 2025; Singh, 2024) pointed out that I need to remove at least one variable and try Ridge or Lasso transformation. After reading Wohlwend's (2023) article, I opted to use Ridge regression as an alternative approach.

For the second part of my question, I created the model based on the book *Mathematical Modeling* (Meerschaert, 2013), BasuMallick's (2022) article and Labonne's (2022) article. As I read more about the linear programming model to better understand it, given my lack of prior experience, I realized my historical data won't be sufficient to determine an optimized energy mix. Therefore, I decided to use the life-cycle assessment (LCA) for each energy source that I found previously. Upon conducting additional research, I uncovered inconsistencies within the data obtained from various trustworthy sources; therefore, to maintain the integrity of my analysis, I used averages or midpoints where applicable (see Table 1).

Table 1

Life-cycle assessments for each energy source, from extraction to disposal in tCO_2e/TWh

Energy Source	**LCA without CCS (tCO_2e/TWh)**	**LCA with CCS (tCO_2e/TWh)**
Hydropower	24000	24000
Wind	12000	12000
Nuclear	8000	8000
Solar	48000	48000
Natural Gas	430000	130000
Oil	640000	224000
Coal	751000	280000
Biofuels	50000	50000
Other Renewables	44000	44000

Note. CCS = carbon capture and storage
I derived the life-cycle assessment (LCA) for each energy source, measured in tCO_2e/TWh, from extraction to disposal, by analyzing data from several sources (Schlömer et al, 2014; Solaris Renewables, 2022; UNECE, 2022; World Nuclear Association, 2011).

The initial model run, using all coefficients, and no constraints yielded an optimal solution: 100% nuclear energy, meeting demand and emission targets. This unrealistic result was expected, as I didn't indicate any bounds for each energy source or constraints, and the model prioritized the energy source with the lower LCA (nuclear energy). The model's bounds are detailed in Table D3 (Appendix D). To produce more realistic outcomes, I also introduced additional constraints, including a minimum baseload requirement of 50% for grid stability, as well as upper limits for solar and wind due to intermittency factors such as the absence of sunlight at night and variability in wind availability (NPUC, 2023; United States Environmental Protection Agency, 2022).

Tools

In terms of the tools employed, since the discovered data was formatted as CSV files, I used Microsoft Excel to merge, refine, and ultimately prepare my dataset for use with the regression model. To help me with the correct regression model choice, I used analytical tools such as VIF (Variance Inflation Factor) and correlation matrices for multicollinearity and Ramsey RESET test and scatterplots for linearity, as I mentioned in the Selection section.

For my data analysis, I chose Python. I installed Visual Studio Code, Jupyter notebook, Python and all the recommended libraries for data analysis, such as NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Statsmodel, and Pulp (Ali, 2024). To configure everything, I followed a video tutorial titled *Python for Data Analysts and Data Scientists* by Shashank Kalanithi (2021). The code for the Ridge regression model and the LP model are attached as Appendix B and Appendix C, respectively.

Analysis

The Ridge regression model using seven predictors (percentages of solar, wind, hydropower, nuclear, natural gas, oil, and coal) and emissions intensity (tCO_2e/TWh) as the response variable, achieved an R^2 score of 0.9878, indicating an excellent fit to historical U.S. data (1965–2023). In other words, the model explains 98.78% of the variation in emissions intensity, leaving only about 1.22% as unexplained noise from random fluctuations, omitted factors, or measurement error. The output for the Ridge regression model in Python, obtained using the *scikit-learn* library can be seen in Figure 1.

Figure 1

Ridge Regression model output

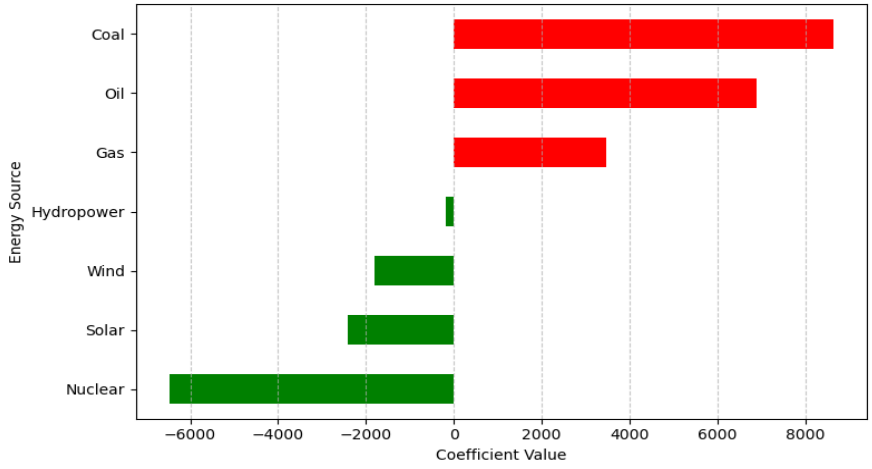
```
R² Score: 0.9878
MSE: 4014341.9033
RMSE: 2003.5823
Best alpha (λ): 0.08697490026177834
Ridge Regression Coefficients:
Solar_Percentage        -2402.634895
Wind_Percentage         -1811.725124
Hydropower_Percentage    -189.253878
Nuclear_Percentage      -6480.226606
Gas_Percentage           3475.638935
Oil_Percentage           6890.915038
Coal_Percentage          8643.490573
```

The Ridge regression model also achieved a Mean Squared Error (MSE) of 4014342 and a Root Mean Squared Error (RMSE) of 2004 tCO_2e/TWh, indicating that, on average, predictions deviate from actual emissions intensity values by roughly 2,004 tCO_2e/TWh. Given the historical variation in emissions intensity, this represents a relatively small prediction error and supports the model's strong fit to the data. The optimal regularization parameter, λ = 0.08697, determined via cross-validation, reflects a low degree of coefficient shrinkage. This implies that the model needed slight regularization to lessen multicollinearity's impact. However,

the adjustment was small, keeping the coefficients near their original multiple linear regression values, boosting stability, and enhancing prediction accuracy.

Figure 2

Ridge Regression Coefficients (tCO_2e/TWh per One-Standard-Deviation Increase)



The signs and magnitudes of the coefficients (Figure 2) aligned with expectations: renewables and nuclear had large negative coefficients (e.g., nuclear −6480, solar −2403), indicating reductions in emissions intensity, while fossil fuels had large positive coefficients (e.g., coal +8643, oil +6890). These results are consistent with life-cycle emissions data, and the United Nation's (2025) claim that fossils contribute the most to the greenhouse gas emissions, supporting the validity of the model for identifying how changes in energy composition affect emissions intensity. The interpretation of each energy source's coefficient can be seen in Table D1 (Appendix D).

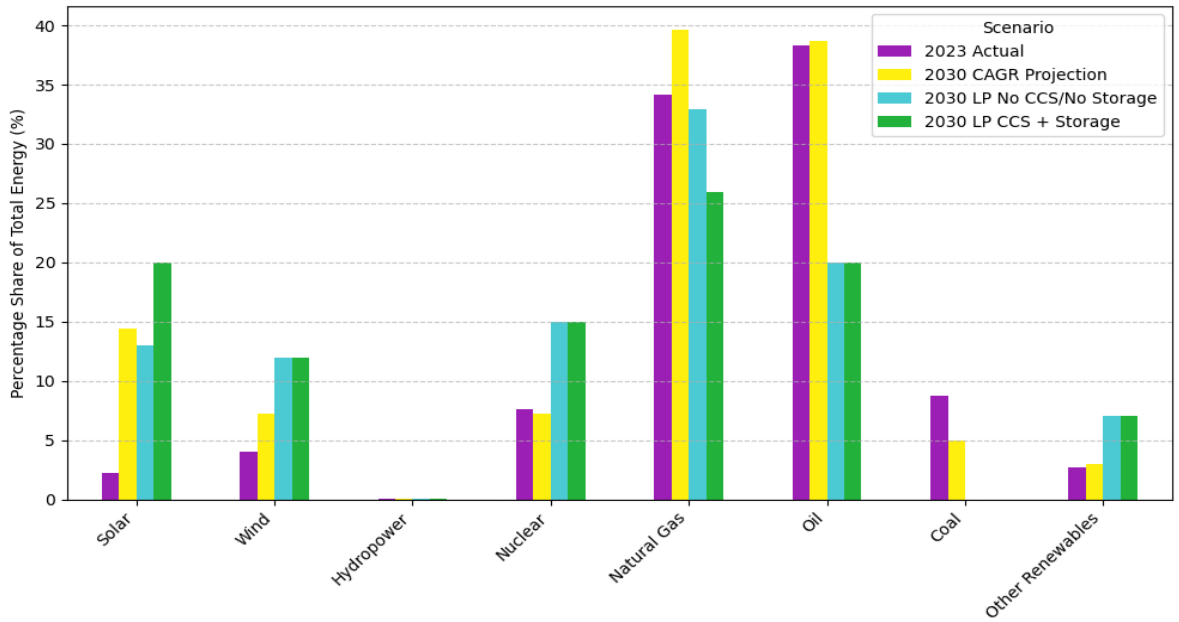
Because the energy-share predictors were standardized before fitting the Ridge regression model, the coefficients represent changes associated with a one-standard-deviation increase in each predictor rather than a one-percentage-point increase. The size of one standard deviation differs among the energy sources because their historical shares varied by different amounts. For example, one standard deviation corresponds to 0.484 percentage points for solar and 3.978

percentage points for coal. The historical means and standard deviations used in the standardization process are presented in Table D2 in Appendix D.

The Linear Programming model, built using life-cycle assessment (LCA) emission factors and realistic capacity bounds derived from historical growth rates (CAGR) and Ridge coefficients (Table D3, Appendix D), was used to find an optimal 2030 energy mix that minimizes total emissions. Under realistic bounds and current LCA values, the model could not achieve the 50 % emissions reduction target, indicating that such a target is unattainable under current trends without significant technological or policy changes. Not even increasing storage capacity for solar and wind energy could achieve this reduction.

Figure 3

Energy Mix Comparison: 2023 vs. 2030 Projections and LP Scenarios



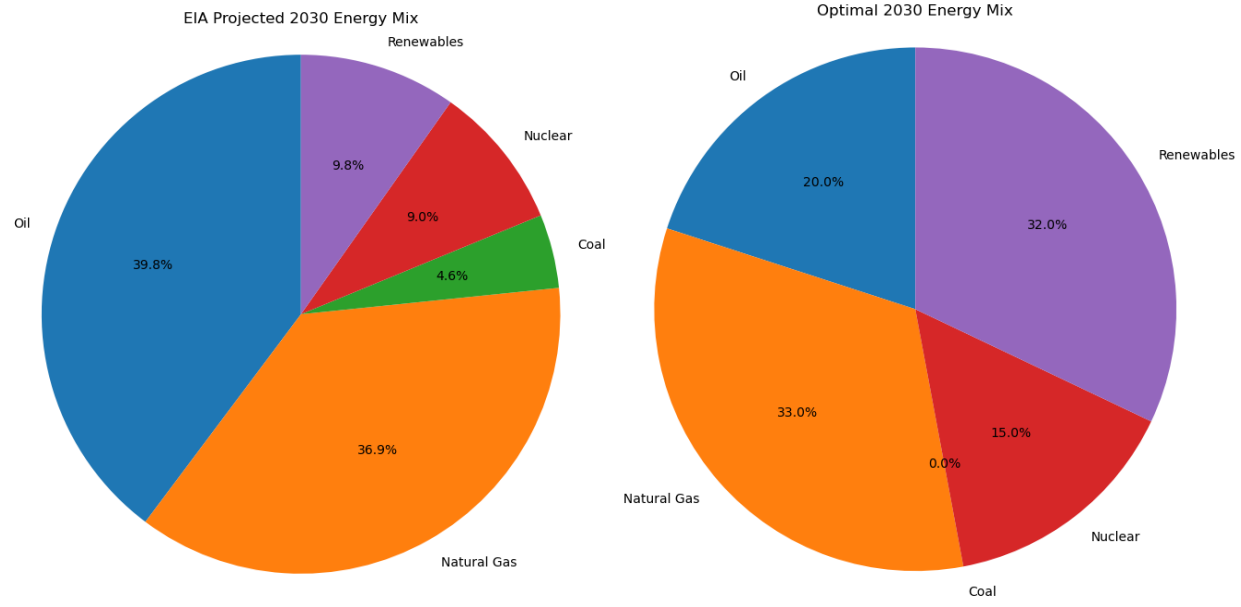
However, scenarios with LCA values reflecting carbon capture and storage (CCS) showed that the target could be met, with total emissions reduced to approximately 2.68 billion tCO_2e. Adding increased storage capacity for wind and solar further lowered emissions to approximately 2.53 billion tCO_2e. The percentages for each energy source are shown in Figure 3,

across different scenarios, including the 2023 composition, the 2030 CAGR-based projections, and the optimal mixes from the LP model.

Compared to the actual 2030 projections from the *Annual Energy Outlook 2025* report by the U.S. Energy Information Administration (EIA, 2025), the optimal mix prioritizes renewable energy, 32% vs. 9.8%, and diminishes the fossil fuels energy by a lot, 53% vs. 81.3% (see Figure 4). It is important to note that the 2030 projections are based on today's regulations and possibilities, while the optimal mix considers the implementation of carbon capture and storage technologies and prioritizes energy sources with lower carbon emissions. Even though achieving this composition is potentially feasible, it is important to recognize that the associated expenses will be exceptionally high.

Figure 4

EIA projected 2030 Energy mix vs. Optimal 2030 Energy Mix (CCS, no increased storage)



Overall, the models fit the question and information excellently. The Ridge regression model quantifies relationships with high accuracy, and the Linear Programming model optimizes

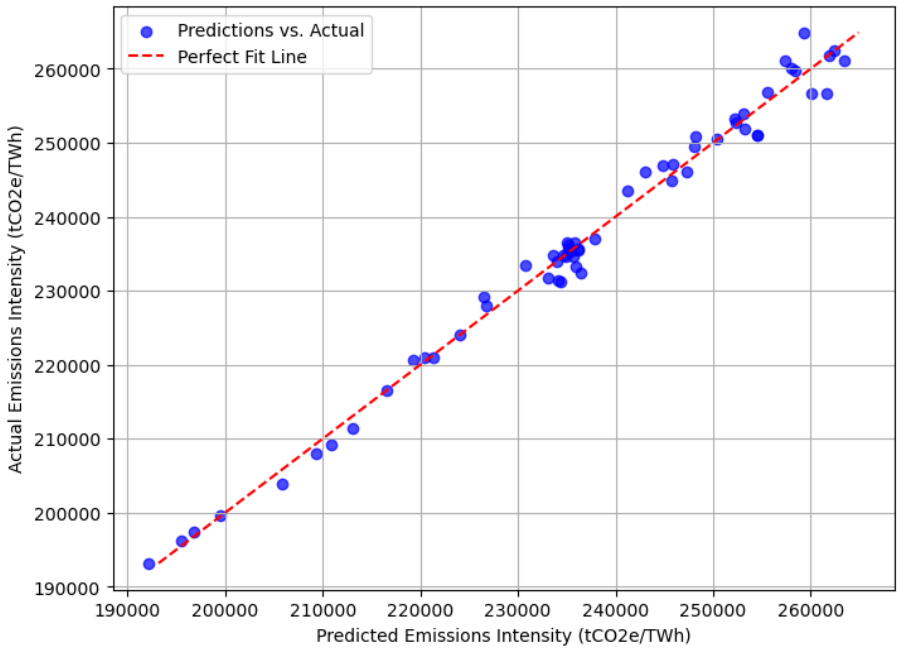
under constraints, though carbon capture and storage technologies are vital enabling components. The results are better than expected in showing the technology needed for decarbonization.

Limitations

Ridge regression ensured stable coefficient estimates given the high correlation between predictors; however, some constraints impact its interpretability and use. First, the model assumes a strictly linear relationship between energy composition and emissions intensity, which may overlook nonlinear dynamics such as threshold effects or diminishing returns from certain technologies. While Ridge regression lessens the effect of multicollinearity, it does not remove it completely, as the predictors' inherent dependency, caused by the percentages totaling 100%, is a structural problem. This limitation is evident in Figure 5, where predicted versus actual emissions intensities show strong alignment overall but also reveal deviations for certain years, suggesting that omitted nonlinear effects or interactions may still influence the results.

Figure 5

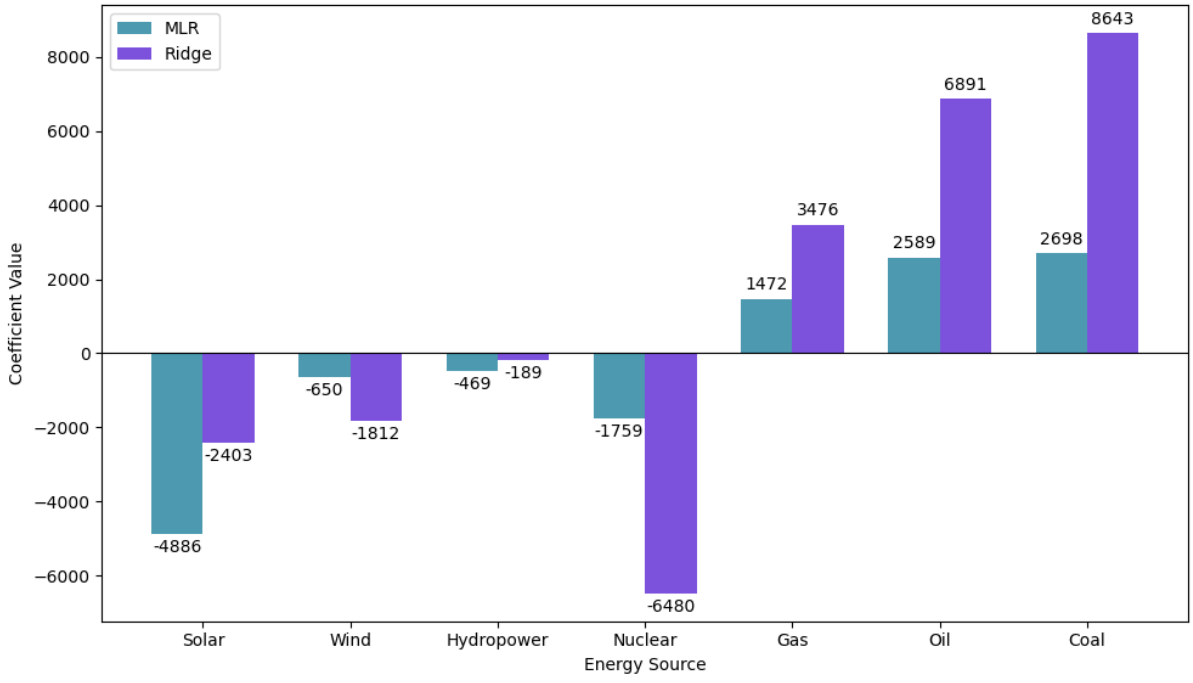
Predicted vs. Actual Emissions Intensity from Ridge Regression Model



Also, the model does not generate typical OLS statistics such as p-values or confidence intervals, which makes it difficult to assess the statistical significance of the coefficients. The elimination of "Other Renewables" and "Biofuels" predictors further limit practical interpretation, preventing direct estimation of their individual contributions. Moving from Multiple Linear Regression to Ridge Regression causes a significant shift in the coefficients and may introduce bias. To illustrate this, Figure 6 compares coefficients from the MLR and Ridge models. The degree of change, especially concerning coal, oil, and nuclear energy, shows how predictor correlation skewed MLR's estimates. Ridge corrects this instability, but at the cost of introducing bias and reducing interpretability of the exact coefficient values.

Figure 6

Comparison of MLR and Ridge Coefficients



The model is also sensitive to alpha. As Table 2 shows, higher alpha values shrink coefficients more significantly, such as reducing the gas coefficient from 4870.6 at alpha 0.01 to -308.9 at alpha 100, while maintaining a high R^2 score above 0.847.

Table 2

Ridge Regression Coefficients and R^2 Scores Across Different Alpha Values

Alpha (λ)	Natural Gas Coefficient	Nuclear Coefficient	Solar Coefficient	R^2 Score
0.010000	4870.6	-5430.0	-2372.0	0.988
0.215443	2418.2	-7238.0	-2435.0	0.988
0.599484	1426.9	-7824.0	-2511.0	0.987
1.668101	932.0	-7791.0	-2702.0	0.987
4.641589	704.6	-7194.0	-3059.0	0.985
12.915497	420.3	-6244.0	-3347.0	0.978
35.938137	20.2	-5028.0	-3226.0	0.950
100.000000	-308.9	-3513.0	-2613.0	0.847

Note. Alpha (λ) is the regularization parameter. It is called lambda in theory and alpha in Python as lambda is reserved by the Python language.

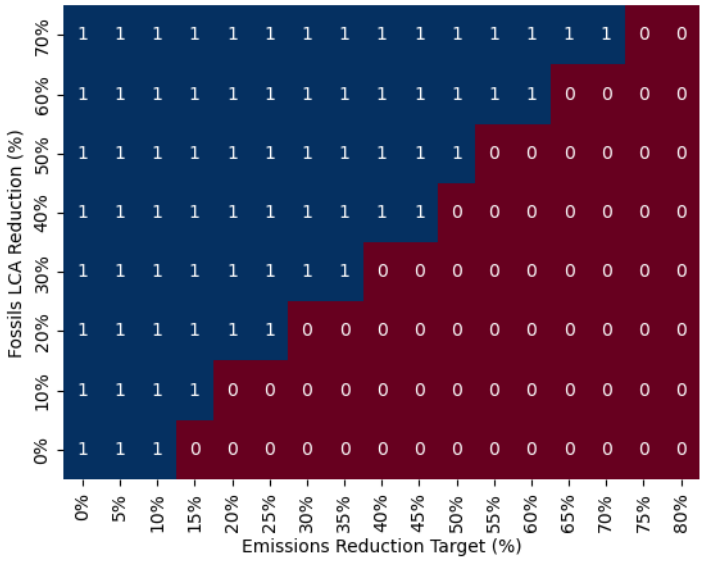
The Linear Programming (LP) model also faces significant limitations. It relies on static life-cycle assessment (LCA) emission factors, assuming no improvements in technology efficiency between 2023 and 2030. Because these LCA values are usually presented as ranges instead of precise figures, they possess some degree of uncertainty. Furthermore, the LP model only aims to minimize emissions, disregarding costs, political practicality, and infrastructure constraints. The model's feasibility depends entirely on the defined constraints, shown in Figure E1 (Appendix E). Because these rules are interdependent, even minor adjustments can render the model infeasible, as shown in sensitivity tests.

Model infeasibility was a recurring challenge, as indicated by the solver output in Table E1 (Appendix E), where certain constraint combinations prevented any solution. Sensitivity testing revealed that the emissions cap was a primary cause of infeasibility when fossil fuel

LCAs without carbon capture and storage (CCS) were used. Transitioning to LCAs with CCS provides workable solutions, but it operates on the assumption that widespread CCS deployment is technologically and economically achievable within the specified timeframe. As shown in Figure 7, the current LCA can reduce emissions by 10%; however, for every additional 5%-10% reduction in emissions, a 10% reduction in fossil LCA is necessary.

Figure 7

Feasibility Heatmap: Fossils LCA Reduction (%) vs. Emissions Reduction Target (%)

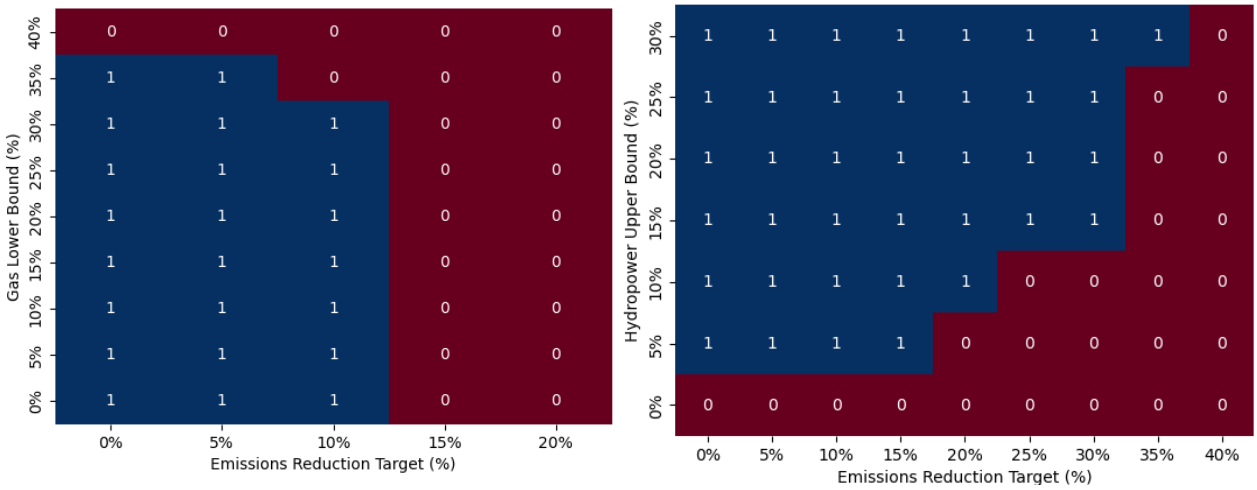


Note. 0-infeasible, 1-feasible.

The LP model is also sensitive to lower and upper bounds. The feasibility heatmaps in Figure 8 illustrate this for natural gas lower bounds and hydropower upper bounds, both part of baseload. Natural gas's high lifecycle assessment means the model can't achieve emissions reductions >10% with lowers >35%, as it forces more high-emission energy; conversely, hydropower's low LCA could enable up to 35% reductions with higher uppers, but geographic limits cap its bound at 0.05%, restricting impact. More feasibility maps are included in Appendix E (Figure E2-E5).

Figure 8

Feasibility Heatmaps: Gas Lower Bound (%) vs. Emissions Reduction Target (%) and Hydropower Upper Bound (%) vs. Emissions Reduction Target (%)



Note. 0-infeasible, 1-feasible.

Approach

To answer my research question, I followed the five-step method suggested by the textbook, *Mathematical Modelling* (Meerschaert, 2013): ask the question, select the modeling approach, formulate the model, solve the model and answer the question. Once I decided on a topic, I created a question that was relevant to the topic and feasible to answer within the duration of the course. With the question in hand, and based on experience from other applied mathematics classes, I anticipated needing a regression model. I therefore located historical data to examine the relationship between each energy source and GHG emissions over time. Initially, I planned to use only a multiple linear regression (MLR) model to determine these relationships and then apply the results to find the optimal mix. However, further research showed that I would need two separate models—a regression model and an optimization model—and that the regression results could not be used directly as emission factors in the optimization stage. My final decision was to use a Ridge regression model and a Linear Programming (LP) model, the

reasons for which are detailed in the Selection section. Once I learned more about Linear Programming, I decided to use the regression coefficients and the compound annual growth rate of each energy source to set capacity bounds in the LP model. During the "formulate the model" step, after creating the dataset for the regression model, I created and coded both models in Python. Details on building the models are discussed in the Creation, Process and Tools sections. To solve the model, I ran both models in Python and performed several sensitivity tests, which are reviewed in the Limitations section.

With the final models in place, I was able to answer the first part of my question using the Ridge regression coefficients. However, the LP model initially returned "Infeasible," meaning no optimal mix could be found under the given constraints and bounds. I experimented with adjusting other constraints, such as the baseload minimum and intermittency maximum, but this still produced no feasible solution. The model only worked after removing the emissions cap constraint, but that scenario resulted in very high total emissions. The issue was that the life-cycle assessment (LCA) emissions for fossil fuels were too high to allow a 50% reduction under current assumptions. I then remembered that LCA values for fossil fuels are typically presented in two ranges: one excluding carbon capture and storage (CCS) and another including it. When I replaced the fossil fuel LCAs with the CCS-adjusted values, the model produced an optimal solution instead of being infeasible. Therefore, to answer the second part of my question, I used the LCA values with CCS and explained this choice in the Analysis section.

Applicability

The primary purpose of my models was to address a two-part research question: first, to determine the statistical relationship between the composition of the U.S. energy mix and greenhouse gas (GHG) emissions intensity, and second, to identify an optimal future mix that

could minimize emissions under realistic constraints. Overall, the models are highly applicable to the research question and align closely with their objectives. The Ridge regression model worked well for the first part because it reduced instability from multicollinearity in percentage-based energy data, leading to more reliable coefficient estimates compared to a traditional multiple linear regression model. These coefficients showed that coal, natural gas, and oil had the strongest positive correlation with emissions intensity, whereas nuclear, hydropower, wind, and solar power were most strongly associated with emissions reductions.

The linear programming (LP) model addressed the second part of the research question by using realistic bounds informed by historical growth rates (CAGR) and the Ridge regression results. Using standard life-cycle assessment (LCA) values, the LP model produced infeasible results, indicating that no energy mix could satisfy the constraints while reducing emissions by 50%. However, when the LCA values for fossil fuels were adjusted to reflect carbon capture and storage (CCS) technologies, the model yielded an optimal mix that met the emissions target: Solar 13%, Wind 12%, Hydropower 0.05%, Nuclear 15%, Natural Gas 32.95%, Oil 20%, Coal 0%, Other Renewables 7%, resulting in 2.68 $GtCO_2e$ total emissions — a 68.6% reduction from 2005 levels. Therefore, an optimal 2030 energy mix that meets a 50% emissions reduction target is only achievable under current constraints if fossil fuel generation incorporates large-scale CCS adoption.

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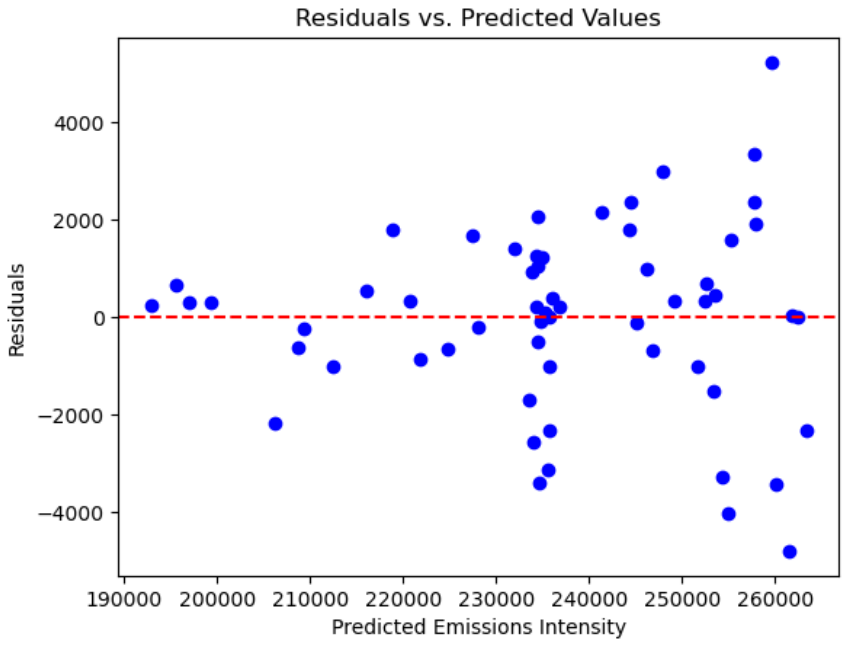
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Appendix A

Supplemental tables and figures

Figure A1

Residuals vs. Predicted Values Scatterplot to confirm linearity



Note. The plot shows random scatter around zero with no clear patterns. The residuals are evenly distributed with no apparent clustering or trends, suggesting the linear assumption is valid.

Table A1

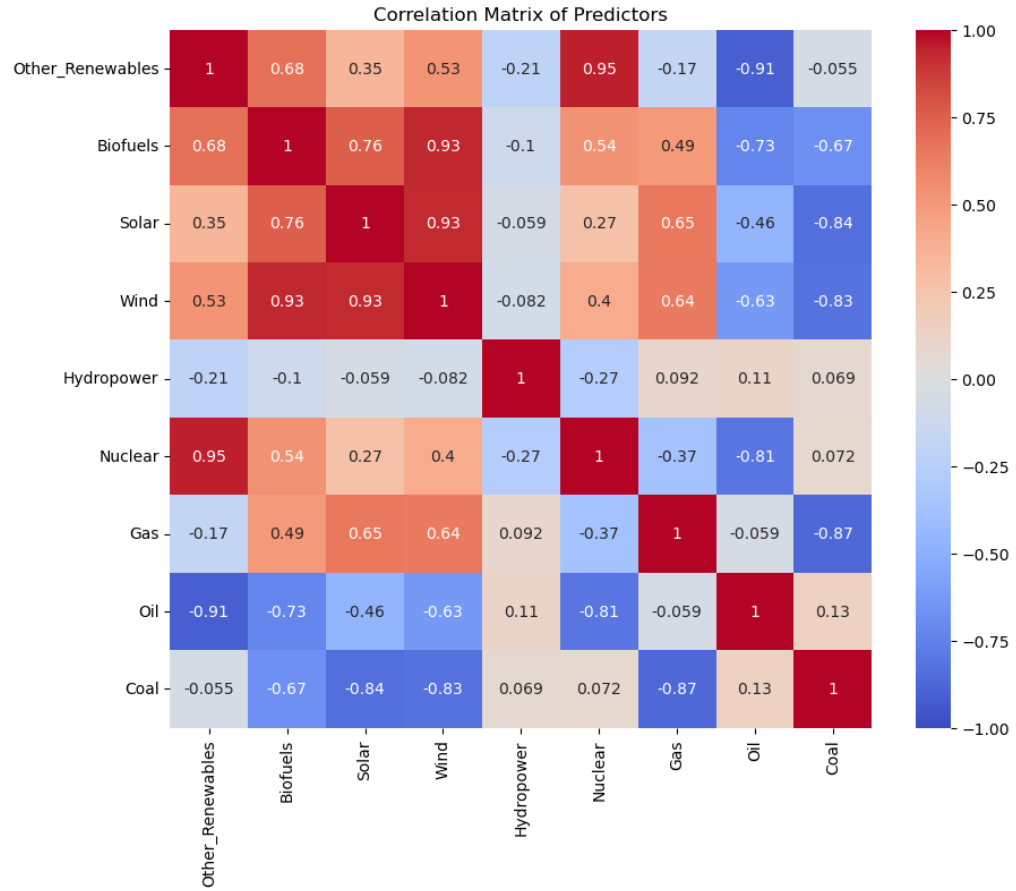
Variance Inflation Factor (VIF) values for each predictor

Predictor	VIF With 9 predictors	VIF With 7 predictors
Other Renewables	359.21	-
Biofuels	49.20	-
Solar	32.73	10.27
Wind	137.50	34.28
Hydropower	1.30	1.27
Nuclear	223.61	32.03
Gas	457.98	251.57
Oil	389.35	234.01
Coal	186.85	175.56

Note. A VIF value greater than 10 generally indicates high multicollinearity; all predictors except Hydropower far exceed this threshold, confirming the presence of strong multicollinearity in the dataset. Even after removing two predictors, multicollinearity still persists.

Figure A2

Correlation matrix of predictors



Note. The correlation matrix shows high positive correlation between Biofuels and Wind, Solar and Wind, Other Renewables and Nuclear.

Appendix B

Python Code for the Ridge Regression Model

```
import pandas as pd
import numpy as np
from sklearn.linear_model import Ridge
from sklearn.model_selection import cross_val_score, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, r2_score
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv(r"C:\path\to\1965_2023_Data_USA.csv")

# Define independent and dependent variables
X = df[['Solar_Percentage',
              'Wind_Percentage', 'Hydropower_Percentage', 'Nuclear_Percentage',
              'Gas_Percentage', 'Oil_Percentage', 'Coal_Percentage']]
y = df['Emissions_Intensity']  # dependent variable

# Standardize the predictors
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Perform Ridge regression with hyperparameter tuning
alphas = np.logspace(-3, 5, 100)  # λ values to search

ridge = Ridge()
params = {'alpha': alphas}
ridge_cv = GridSearchCV(ridge, params, cv=5, scoring='neg_mean_squared_error')
ridge_cv.fit(X_scaled, y)

# Best alpha and fit model
best_alpha = ridge_cv.best_params_['alpha']
ridge_best = Ridge(alpha=best_alpha)
ridge_best.fit(X_scaled, y)

y_pred = ridge_best.predict(X_scaled)

r2 = r2_score(y, y_pred)
mse = mean_squared_error(y, y_pred)
rmse = np.sqrt(mse)
```

```
print(f"R² Score: {r2:.4f}")
print(f"MSE: {mse:.4f}")
print(f"RMSE: {rmse:.4f}")

# Show coefficients
coefficients = pd.Series(ridge_best.coef_, index=X.columns)
print("Best alpha (λ):", best_alpha)
print("Ridge Regression Coefficients:")
print(coefficients)

# Plot coefficients
coefficients = coefficients.sort_values()

coefficients = coefficients.rename(index={
    'Solar_Percentage': 'Solar',
    'Wind_Percentage': 'Wind',
    'Hydropower_Percentage': 'Hydropower',
    'Nuclear_Percentage': 'Nuclear',
    'Gas_Percentage': 'Natural Gas',
    'Oil_Percentage': 'Oil',
    'Coal_Percentage': 'Coal'
})

plt.figure(figsize=(8, 5))
coefficients.plot( kind='barh', color=['green' if value < 0 else 'red' for value
in coefficients])
plt.title('Ridge Regression Coefficients (tCO₂e/TWh per 1% Share Change)')
plt.xlabel('Coefficient Value')
plt.ylabel('Energy Source')
plt.grid(True, axis='x', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

Appendix C

Python Code for the Linear Programming Model

```
import pulp

# Define problem
prob = pulp.LpProblem("Min_Emissions_Energy_Mix", pulp.LpMinimize)

# Variables: % shares (0-1)
sources = ['Solar', 'Wind', 'Hydropower', 'Nuclear', 'Natural_Gas', 'Oil',
'Coal', 'Other_Renewables']
P = pulp.LpVariable.dicts("P", sources, lowBound=0, upBound=1)

# LCA coefficients (tCO₂e/TWh)
LCA = {
    'Solar': 48000, 'Wind': 12000, 'Hydropower': 24000,
    'Nuclear': 8000, 'Natural_Gas': 130000, 'Oil': 224000, 'Coal': 280000,
    'Other_Renewables': 44000
}

demand = 26921  # TWh

# Objective
prob += pulp.lpSum([P[s] * demand * LCA[s] for s in sources])

# Constraints
prob += pulp.lpSum([P[s] for s in sources]) == 1, "Sum_100"

bounds_lower = {'Solar': 0.01, 'Wind': 0.05, 'Hydropower': 0.0002, 'Nuclear':
0.08, 'Natural_Gas': 0.15, 'Oil': 0.20, 'Coal': 0.0, 'Other_Renewables': 0.03}
bounds_upper = {'Solar': 0.20, 'Wind': 0.12, 'Hydropower': 0.0005, 'Nuclear':
0.15, 'Natural_Gas': 0.40, 'Oil': 0.30, 'Coal': 0.04, 'Other_Renewables': 0.07}
for s in sources:
    prob += P[s] >= bounds_lower[s], f"Lower_{s}"
    prob += P[s] <= bounds_upper[s], f"Upper_{s}"

prob += pulp.lpSum([P[s] * demand * LCA[s] for s in sources]) <=  4265000000,
"Emissions_Cap"

prob += P['Nuclear'] + P['Hydropower'] + P['Wind'] + P['Coal'] + P['Natural_Gas']
>= 0.50, "Baseload_Min"

prob += P['Solar'] + P['Wind'] <= 0.25, "Intermittency_Max"
```

```
prob += P['Coal'] + P['Oil'] + P['Natural_Gas'] <= 0.55, "Fossil_Cap"

# Solve
prob.solve()

# Check feasibility
status = pulp.LpStatus[prob.status]
print(f"Model Status: {status}")

# If feasible, print results
if status == 'Optimal':
    print("Optimal Energy Mix (%):")
    for s in sources:
        print(f"{s}: {P[s].varValue * 100:.2f}%")
    print(f"Total Emissions: {pulp.value(prob.objective):,.0f} tCO₂e")
else:
    print("No optimal solution found. Check constraints for conflicts.")
```

Appendix D

Supplemental tables for the Analysis Section

Table D1

The interpretation of the Ridge coefficients

Energy Source	Coefficient (rounded)	Interpretation
Solar (%)	–2,403	+1SD solar in solar share → –2,403 tCO_2e/TWh emissions intensity
Wind (%)	–1,812	+1SD wind → –1,812 tCO_2e/TWh emissions intensity
Hydropower (%)	–189	+1SD hydropower → –189 tCO_2e/TWh emissions intensity
Nuclear (%)	–6,480	+1SD nuclear → –6,480 tCO_2e/TWh emissions intensity
Natural Gas (%)	+3,476	+1SD gas → +3,476 tCO_2e/TWh emissions intensity
Oil (%)	+6,891	+1SD oil → +6,891 tCO_2e/TWh emissions intensity
Coal (%)	+8,643	+1SD coal → +8,643 tCO_2e/TWh emissions intensity

Note. SD = standard deviation. Because the predictors were standardized before fitting the Ridge model, each coefficient represents the predicted change in emissions intensity associated with a one-standard-deviation increase in the corresponding energy source's historical share. The coefficients describe model-based associations and should not be interpreted as causal effects.

Table D2

Historical Means and Standard Deviations of the Ridge Regression Predictor

Energy Source	Historical Mean Share	Standard Deviation
Solar	0.194%	0.484 percentage points
Wind	0.634%	1.131 percentage points
Hydropower	0.098%	0.491 percentage points
Nuclear	5.954%	2.966 percentage points
Natural Gas	26.399%	3.497 percentage points
Oil	41.969%	3.258 percentage points
Coal	20.405%	3.978 percentage points

Note. Values are based on annual U.S. energy-source shares from 1965 to 2023. The historical mean is the average share of each source during this period. The standard deviation measures the typical variation in each source's historical share and was used to standardize the predictors before fitting the Ridge regression model.

Table D3

Bounds for the LP model using CAGR and Ridge Coefficients

Energy Source	2023%	CAGR (2008-2023)(%)	CAGR (2013-2023)(%)	CAGR (2018-2023)(%)	Predicted 2030, CAGR 10 years	Ridge Coeff.	Priority	Lower Bound	Upper Bound
Solar	2.261	38.889	30.322	20.742	14.435	-2403	2	10	20
Wind	3.985	13.965	8.9374	9.1427	7.256	-1812	3	5	12
Hydropower	0.023	-0.602	-1.591	-3.19	0.021	-189	4	0.02	0.05
Nuclear	7.641	-0.717	-0.818	-0.757	7.214	-6480	1	8	15
Gas	34.12	2.4009	2.1413	2.024	39.578	3476	6	15	40
Oil	38.29	-0.133	0.1812	-0.214	38.773	6891	7	20	30
Coal	8.739	-6.433	-7.765	-8.807	4.963	8643	8	0	4
Other Renewables (geothermal, biofuels, other)	2.705	2.9955	1.3224	1.746	2.966		5	3	7

Note. CAGR = Compound Annual Growth Rate

Appendix E

Supplemental figures and tables for the Limitations Section

Table E1

Limitations of the LP model with different scenarios

Scenario	PulP Output	Limitations
• No CCS • No CCS + Storage (10% increase in intermittency max)	`Model Status: Infeasible`	The model is infeasible because it can't achieve a 50% emission cap. LCA values dominate the fossil shares required by bounds (gas 34.95%, oil 20%), making it impossible to meet the cap without adjustments. This shows the limitation of assuming no mitigation technologies, as the constraints force reliance on high-LCA sources.

CCS (~65-70% reduction in LCA)	`Model Status: Optimal`
	`Optimal Energy Mix (%):`
	`Solar: 13.00%`
	`Wind: 12.00%`
	`Hydropower: 0.05%`
	`Nuclear: 15.00%`
	`Natural_Gas: 32.95%`
	`Oil: 20.00%`
	`Coal: 0.00%`
	`Other_Renewables: 7.00%`
	`Total Emissions: 2,681,520,047 tCO2e`
		The model assumes aggressive reductions in fossil LCAs which makes the cap achievable but limits realism if deployment doesn't scale.

CCS (~65-70% reduction in LCA)
+ Storage (10% increase in intermittency max)
	`Model Status: Optimal`
	`Optimal Energy Mix (%):`
	`Solar: 20.00%`
	`Wind: 12.00%`
	`Hydropower: 0.05%`
	`Nuclear: 15.00%`
	`Natural_Gas: 25.95%`
	`Oil: 20.00%`
	`Coal: 0.00%`
	`Other_Renewables: 7.00%`
	`Total Emissions: 2,526,993,507 tCO2e`
		Combining LCA reductions with higher intermittency boosts renewables (solar 20%), lowering emissions further, but the model ignores interactions like storage efficiency losses.

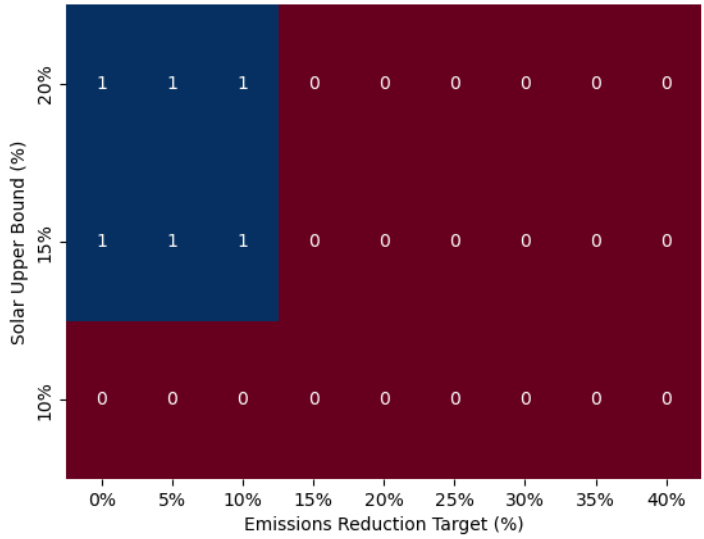
Figure E1

Python Implementation of LP constraints

```
# Constraints
prob += pulp.lpSum([P[s] for s in sources]) == 1, "Sum_100"
bounds_lower = {'Solar': 0.01, 'Wind': 0.05, 'Hydropower': 0.0002, 'Nuclear':
0.08, 'Natural_Gas': 0.15, 'Oil': 0.20, 'Coal': 0.0, 'Other_Renewables': 0.03}
bounds_upper = {'Solar': 0.20, 'Wind': 0.12, 'Hydropower': 0.0005, 'Nuclear':
0.15, 'Natural_Gas': 0.40, 'Oil': 0.30, 'Coal': 0.04, 'Other_Renewables':
0.07}
for s in sources:
    prob += P[s] >= bounds_lower[s], f"Lower_{s}"
    prob += P[s] <= bounds_upper[s], f"Upper_{s}"
prob += pulp.lpSum([P[s] * demand * LCA[s] for s in sources]) <= 4265000000,
"Emissions_Cap"
prob += P['Nuclear'] + P['Hydropower'] + P['Wind'] + P['Coal'] +
P['Natural_Gas'] >= 0.50, "Baseload_Min"
prob += P['Solar'] + P['Wind'] <= 0.25, "Intermittency_Max"
prob += P['Coal'] + P['Oil'] + P['Natural_Gas'] <= 0.55, "Fossil_Cap"
```

Figure E2

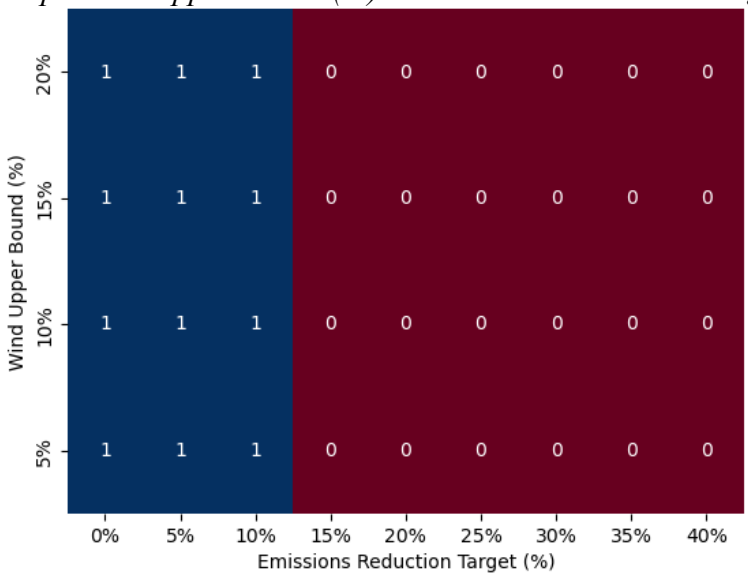
Feasibility Heatmap: Solar Upper Bound (%) vs. Emissions Reduction Target (%)



Note. 0-infeasible, 1-feasible.

Figure E3

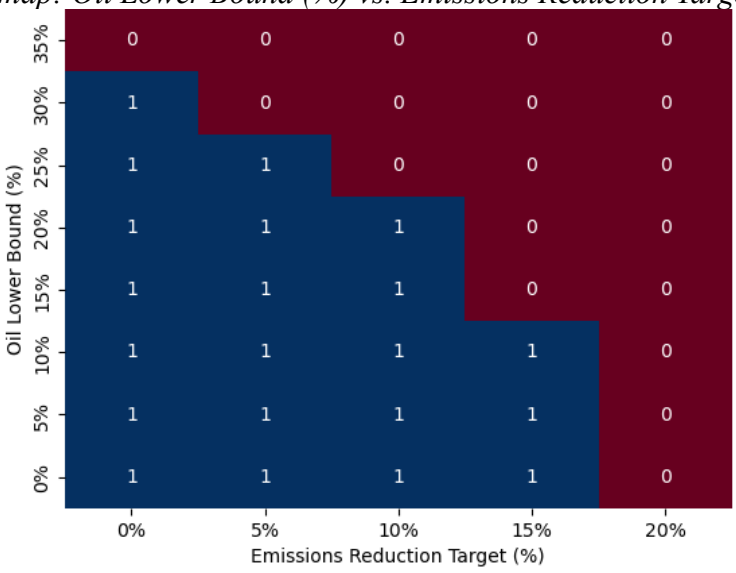
Feasibility Heatmap: Wind Upper Bound (%) vs. Emissions Reduction Target (%)



Note. 0-infeasible, 1-feasible.

Figure E4

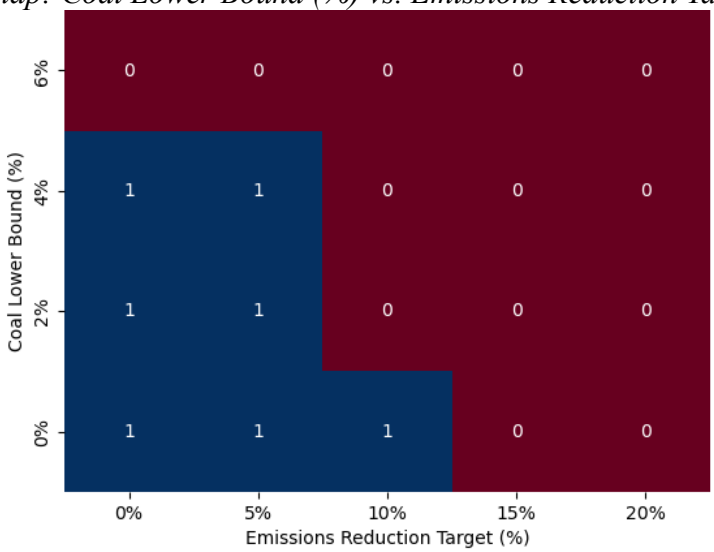
Feasibility Heatmap: Oil Lower Bound (%) vs. Emissions Reduction Target (%)



Note. 0-infeasible, 1-feasible.

Figure E5

Feasibility Heatmap: Coal Lower Bound (%) vs. Emissions Reduction Target (%)



Note. 0-infeasible, 1-feasible.